KS3 DIGITAL TECHNOLOGY

MICROCHIPS AND SEMICONDUCTORS

The tiny switches that run the modern world: what semiconductors are, how transistors make logic gates, how chips are made from sand, what makes a computer fast, why phones have special AI chips, and why countries compete to make them.

MR BUTLER'S DIGITAL TECHNOLOGY

LESSONS

1 Switches made of sand
2 Logic from switches
3 From sand to chip
4 What makes a computer fast?
5 CPUs, GPUs and AI chips
6 The global chip race

LESSON 1 OF 6

SWITCHES MADE OF SAND

Conductors let electricity flow and insulators don't; semiconductors such as silicon can be controlled to do either.

MICROCHIPS AND SEMICONDUCTORS

STARTER · 5 MINUTES

DO NOW

Name something that conducts electricity and something that doesn't. What would be useful about a material that could do both?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain what a semiconductor is and how a transistor works as a switch

SUCCESS CRITERIA: I CAN...

- ✓ I can describe conductors, insulators and semiconductors
- ✓ I can explain why silicon is used for chips
- ✓ I can explain how a transistor works as a switch

THREE KINDS OF MATERIAL

- Conductors let electricity flow easily: copper, gold
- Insulators block it: plastic, rubber, glass
- Semiconductors are in between: we can control when they conduct
- Silicon is the most widely used semiconductor

KEY WORDS

SEMICONDUCTOR	A material, such as silicon, whose ability to conduct electricity can be controlled.
SILICON	The element (found in sand) used to make most computer chips.
TRANSISTOR	A tiny electronic switch made from semiconductor material that represents 1 (on) or 0 (off).
MICROCHIP	A small piece of silicon containing a complete electronic circuit, often with billions of transistors.

SILICON: PURIFIED SAND

- Silicon is the second most common element in the Earth's crust
- Found in sand as silicon dioxide (quartz)
- Purified to 99.9999999% pure for chips
- Small amounts of other elements are added (doping) to control how it conducts

ACTIVITY · 20 MINUTES · PAIRS

TRANSISTOR SWITCH

1. Build the circuit on the circuit simulator (or with the kit) your teacher shows: a battery, a transistor, an LED and a push-button connected to the transistor's gate.
2. Press the button: what happens to the LED? Release it.
3. Record what "1" and "0" mean in your circuit.
4. Explain how millions of these switches could store information.

In your workbook, answer:

? What happened when you pressed and released the button, and what do 1 and 0 represent?

? How could a chip use billions of these switches to store a photo?

Your workbook: Lesson 1 · tick each step off as you go

THE TRANSISTOR

- A transistor is a switch with no moving parts
- A small voltage on its "gate" turns the flow of current on or off
- On = 1, off = 0: transistors store and process bits
- Modern chips contain billions of transistors

QUICK CHECK

1. What is a semiconductor?

A A material that never conducts
B A type of battery
C A material that always conducts
D A material whose conduction can be controlled

2. Which element are most chips made from?

A Plastic
B Copper
C Iron
D Silicon

3. What does a transistor do in a computer?

A Acts as a tiny switch for 1s and 0s
B Displays images
C Cools the chip
D Stores electricity

4. What is doping?

A Cleaning the chip
B Painting the chip
C Testing the chip
D Adding tiny amounts of other elements to silicon

QUICK CHECK: ANSWERS

1. D: A material whose conduction can be controlled

Semiconductors are controllable.

2. D: Silicon

Silicon comes from sand.

3. A: Acts as a tiny switch for 1s and 0s

Transistors switch bits.

4. D: Adding tiny amounts of other elements to silicon

Doping controls conduction.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain the difference between a conductor and a semiconductor.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe conductors, insulators and semiconductors
- ✓ I can explain why silicon is used for chips
- ✓ I can explain how a transistor works as a switch

LESSON 2 OF 6

LOGIC FROM SWITCHES

Combining transistors makes logic gates, which take binary inputs and produce an output.

MICROCHIPS AND SEMICONDUCTORS

STARTER · 5 MINUTES

DO NOW

The school alarm sounds if the fire button is pressed OR smoke is detected. The car starts if the key is in AND the brake is pressed. What's the difference?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe how transistors are combined into AND, OR and NOT logic gates and complete truth tables

SUCCESS CRITERIA: I CAN...

- ✓ I can describe AND, OR and NOT gates
- ✓ I can complete truth tables
- ✓ I can combine gates to solve a simple problem

LOGIC GATES

- A logic gate takes one or more binary inputs and gives one output
- Built from a few transistors
- AND: output 1 only if BOTH inputs are 1
- OR: output 1 if EITHER input is 1
- NOT: flips the input (1 becomes 0)

KEY WORDS

LOGIC GATE	A circuit that takes one or more binary inputs and produces one output, such as AND, OR or NOT.
TRUTH TABLE	A table showing the output of a logic circuit for every possible combination of inputs.
BOOLEAN	Logic that uses only two values: true (1) and false (0).

TRUTH TABLES

- A truth table lists every possible input and the output for each
- Two inputs means 4 rows: 00, 01, 10, 11
- Three inputs means 8 rows
- Engineers use them to check circuits do what they should

GATES EVERYWHERE

- Combine gates to make decisions: "alarm ON if door open AND alarm set"
- Thousands of gates make an adder that can do sums
- Billions make a processor
- The same logic appears in programming: if (a && b)

ACTIVITY · 25 MINUTES · PAIRS

LOGIC GATE CHALLENGES

1. Use the logic gate simulator your teacher shows (e.g. an online circuit builder).
2. Build and test AND, OR and NOT gates, completing a truth table for each.
3. Build the smart alarm: ALARM = SET AND (DOOR OR WINDOW). Test all 8 input combinations.
4. Extension: design a circuit for "the light comes on if it's dark AND movement is detected".

In your workbook, answer:

? Complete the truth table for the smart alarm for these rows: SET=0, DOOR=1, WINDOW=1; SET=1, DOOR=0, WINDOW=0; SET=1, DOOR=1, WINDOW=0.

? Write the logic for the extension and explain it.

Your workbook: Lesson 2 · tick each step off as you go

QUICK CHECK

1. When does an AND gate output 1?

A When either input is 1
B Only when both inputs are 1
C When both inputs are 0
D Never

2. When does an OR gate output 1?

A Only when both inputs are 1
B Only when both are 0
C When at least one input is 1
D Always

3. What does a NOT gate do?

A Always outputs 1
B Stores a bit
C Flips the input: 1 becomes 0 and 0 becomes 1
D Adds two inputs

4. How many rows does a truth table for 3 inputs have?

A 6
B 9
C 8
D 3

QUICK CHECK: ANSWERS

1. B: Only when both inputs are 1

AND needs both.

2. C: When at least one input is 1

OR needs either.

3. C: Flips the input: 1 becomes 0 and 0 becomes 1

NOT inverts.

4. C: 8

$2^3 = 8$ combinations.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Complete the truth table for A OR B.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe AND, OR and NOT gates
- ✓ I can complete truth tables
- ✓ I can combine gates to solve a simple problem

LESSON 3 OF 6

FROM SAND TO CHIP

Chips are made in ultra-clean factories called fabs.

MICROCHIPS AND SEMICONDUCTORS

STARTER · 5 MINUTES

DO NOW

How small do you think the smallest parts of a modern chip are? Smaller than a hair? A blood cell? A virus?

LEARNING INTENTIONS

WE ARE LEARNING TO

Describe the main stages of microchip manufacture

SUCCESS CRITERIA: I CAN...

- ✓ I can describe the stages of making a chip
- ✓ I can explain what photolithography is
- ✓ I can explain why chips are made in clean rooms

STEP BY STEP

- 1 Purify silicon from sand and grow a giant single crystal (an ingot)
- 2 Slice it into thin, polished wafers
- 3 Print circuit patterns onto the wafer with light (photolithography), layer by layer
- 4 Test every chip, cut the wafer up, package each chip

KEY WORDS

WAFER	A thin, polished disc of silicon on which many chips are made at once.
PHOTOLITHOGRAPHY	Printing circuit patterns onto a wafer using light shone through a patterned mask.
CLEAN ROOM	A room where the air is filtered to remove almost all dust, used for making chips.
FAB	A chip fabrication plant: a factory that manufactures microchips.
NANOMETRE	One billionth of a metre.

PRINTING WITH LIGHT

- Photolithography works like printing a photo through a stencil
- Light shines through a mask onto light-sensitive coating on the wafer
- The exposed pattern is etched away; the process repeats for dozens of layers
- The most advanced machines use extreme ultraviolet (EUV) light

CLEANER THAN AN OPERATING THEATRE

- A single speck of dust can ruin a chip
- Fabs use clean rooms with filtered air and workers in "bunny suits"
- Building one fab can cost over £10 billion
- Chip features are measured in nanometres (billionths of a metre)

ACTIVITY · 20 MINUTES · PAIRS

CHIP FACTORY FLOWCHART

1. Use the jumbled cards your teacher gives you showing chip-making stages.
2. Put them in order and draw a flowchart with one sentence per stage.
3. Add a "quality check" decision box: what happens to a chip that fails testing?
4. Add two notes explaining why each stage needs extreme precision.

In your workbook, answer:

? List the stages in order.

? Why must chips be made in clean rooms?

Your workbook: Lesson 3 · tick each step off as you go

QUICK CHECK

1. What is a wafer?

A A thin, polished slice of silicon that chips are made on
B A chip package
C A type of transistor
D A biscuit

2. What is photolithography?

A Printing circuit patterns onto a wafer using light
B Testing a chip
C Polishing silicon
D Taking photos of chips

3. Why are chips made in clean rooms?

A Because it's quieter
B To save electricity
C To keep workers warm
D Dust is bigger than chip features and would ruin them

4. What is a nanometre?

A A unit of speed
B A billionth of a metre
C A thousandth of a metre
D A millionth of a gram

QUICK CHECK: ANSWERS

1. A: A thin, polished slice of silicon that chips are made on

Many chips are made on each wafer.

2. A: Printing circuit patterns onto a wafer using light

It's printing with light.

3. D: Dust is bigger than chip features and would ruin them

Contamination ruins chips.

4. B: A billionth of a metre

1 nm = 0.000000001 m.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Describe three stages in the manufacture of a microchip.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe the stages of making a chip
- ✓ I can explain what photolithography is
- ✓ I can explain why chips are made in clean rooms

LESSON 4 OF 6

WHAT MAKES A COMPUTER FAST?

Performance depends on the processor (clock speed, number of cores, cache) and on memory (how much RAM and how fast storage is).

STARTER · 5 MINUTES

DO NOW

Two laptops cost the same. One has a faster processor; the other has twice the RAM. Which would you pick for video editing? For lots of browser tabs?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain how processing power and memory capacity affect a computer's performance

SUCCESS CRITERIA: I CAN...

- ✓ I can explain clock speed, cores and cache
- ✓ I can explain how RAM and storage affect performance
- ✓ I can describe Moore's law

THE PROCESSOR

- Clock speed: instructions processed per second, in gigahertz (GHz); 3 GHz = 3 billion cycles a second
- Cores: separate processors on one chip that work at the same time
- Cache: tiny, super-fast memory inside the processor
- More cores help only if the software can split up its work

KEY WORDS

CLOCK SPEED	The number of cycles a processor carries out per second, measured in gigahertz (GHz).
CORE	A separate processing unit within a processor.
CACHE	Small, very fast memory inside the processor that holds frequently used data.
RAM	Random access memory: the main memory holding programs and data currently in use.
MOORE'S LAW	The observation that the number of transistors on a chip doubles about every two years.

MEMORY AND STORAGE

- RAM holds the programs and data in use right now
- Too little RAM: the computer slows down as it swaps data to storage
- Storage (SSD/hard drive) keeps files when the power is off
- SSDs are much faster than hard drives, so computers start quicker

MOORE'S LAW

- Gordon Moore (1965): the number of transistors on a chip doubles about every two years
- Held for about 50 years: from thousands to tens of billions of transistors
- Now slowing, as transistors approach the size of atoms
- Engineers now stack chips in 3D and design specialised chips instead

ACTIVITY · 20 MINUTES · PAIRS

SPEC DETECTIVE

1. Your teacher gives you the specs of four devices (a gaming PC, a school laptop, a phone and a 10-year-old desktop).

2. Rank them for three tasks: video editing, web browsing with 30 tabs, and a 3D game.

3. Justify each ranking using clock speed, cores, cache, RAM and storage.

In your workbook, answer:

? Which device is best for video editing, and which specifications support your choice?

? Explain why the 10-year-old desktop might feel slow even for web browsing.

Your workbook: Lesson 4 · tick each step off as you go

QUICK CHECK

1. What does a clock speed of 3 GHz mean?

A 3 billion cycles per second
B 3 thousand cycles per second
C 3 programs at once
D 3 GB of memory

2. What is a core?

A The centre of an apple
B A separate processing unit on a chip
C A type of memory
D A storage drive

3. What happens if a computer runs out of RAM?

A It speeds up
B It deletes files
C It slows down as it swaps data to storage
D It switches off

4. What did Moore's law predict?

A Chips would get slower
B Phones would be banned
C Transistor counts on chips would double about every two years
D Computers would get bigger

QUICK CHECK: ANSWERS

1. **A: 3 billion cycles per second**
Giga = billion.

2. **B: A separate processing unit on a chip**
More cores, more parallel tasks.

3. **C: It slows down as it swaps data to storage**
Swapping is slow.

4. **C: Transistor counts on chips would double about every two years**
It described rapid growth.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain how clock speed, number of cores and cache size affect processor performance.

Look back at today's success criteria: can you tick them all?

- ✓ I can explain clock speed, cores and cache
- ✓ I can explain how RAM and storage affect performance
- ✓ I can describe Moore's law

LESSON 5 OF 6

CPUS, GPUS AND AI CHIPS

CPUs are flexible all-rounders; GPUs have thousands of simple cores that are brilliant at doing the same sum many times at once, which suits graphics and AI.

MICROCHIPS AND SEMICONDUCTORS

STARTER · 5 MINUTES

DO NOW

Would you rather have one genius mathematician or 1,000 pupils who can each do simple sums? Which would be quicker at marking 1,000 times-table tests?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Compare CPUs, GPUs and AI accelerators and explain why specialised chips are increasingly used

SUCCESS CRITERIA: I CAN...

- ✓ I can compare a CPU and a GPU
- ✓ I can explain why GPUs are used for AI
- ✓ I can describe a system on a chip

CPU VS GPU

- CPU: a few powerful cores, good at complex, varied tasks one after another
- GPU: thousands of simpler cores doing the same calculation on lots of data at once
- Parallel processing: many calculations at the same time
- Graphics: millions of pixels, same maths for each, so perfect for GPUs

KEY WORDS

CPU	Central processing unit: the general-purpose processor that runs programs.
GPU	Graphics processing unit: a processor with thousands of simple cores that is good at parallel tasks like graphics and AI.
NPU	Neural processing unit: a processor designed to run AI tasks efficiently.
SYSTEM ON A CHIP	A single chip that combines a CPU, GPU and other components.
PARALLEL PROCESSING	Carrying out many calculations at the same time.

CHIPS FOR AI

- Training AI means trillions of simple multiplications
- GPUs turned out to be brilliant at this, so demand for them exploded
- Companies now design AI accelerators specifically for machine learning
- Phones have an NPU (neural processing unit) for on-device AI

ACTIVITY · 20 MINUTES · WHOLE CLASS

WHO'S THE BEST WORKER?

1. Activity 1: one volunteer (the "CPU") solves 30 simple sums alone while the rest of the class (the "GPU") solves them with one sum each. Time both.
2. Activity 2: the CPU solves one multi-step logic puzzle; the GPU class must each do one step without talking to each other. Time both.
3. Record which "processor" won each round and why.

In your workbook, answer:

? Which processor won each round, and why?

? Explain why GPUs are used to train AI models.

Your workbook: Lesson 5 · tick each step off as you go

SYSTEM ON A CHIP

- A phone's main chip combines CPU, GPU, NPU, memory controllers and more
- Called a system on a chip (SoC)
- Saves space and power, which is vital for battery life
- Many are based on designs from Arm, a UK company in Cambridge

QUICK CHECK

1. What is a GPU best at?

A Connecting to Wi-Fi
B Running one complex task step by step
C Storing files
D Doing the same calculation on lots of data at once

2. What is parallel processing?

A Using two screens
B Saving files twice
C Doing many calculations at the same time
D Processing in a straight line

3. What is an NPU?

A A network port
B A neural processing unit designed for AI tasks
C A type of RAM
D A storage drive

4. What is a system on a chip?

A One chip combining CPU, GPU and other parts
B A chip with a tiny operating system
C A chip inside a chip
D A computer case

QUICK CHECK: ANSWERS

1. D: Doing the same calculation on lots of data at once

GPUs excel at parallel work.

2. C: Doing many calculations at the same time

Parallel = simultaneous.

3. B: A neural processing unit designed for AI tasks

NPUs run AI on devices.

4. A: One chip combining CPU, GPU and other parts

SoCs save space and power.

3 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Compare a CPU and a GPU.

Look back at today's success criteria: can you tick them all?

- ✓ I can compare a CPU and a GPU
- ✓ I can explain why GPUs are used for AI
- ✓ I can describe a system on a chip

LESSON 6 OF 6

THE GLOBAL CHIP RACE

Chips are in almost everything, and making them depends on a few companies in a few countries.

MICROCHIPS AND SEMICONDUCTORS

STARTER · 5 MINUTES

DO NOW

In 2021, car factories around the world stopped building cars because of a shortage of something tiny. What was it?

TODAY

LEARNING INTENTIONS

WE ARE LEARNING TO

Explain why semiconductors matter to the economy, national security and the environment

SUCCESS CRITERIA: I CAN...

- ✓ I can describe the global semiconductor supply chain
- ✓ I can explain why governments see chips as a security issue
- ✓ I can describe environmental impacts of chip-making

A GLOBAL SUPPLY CHAIN

- Designers (e.g. Arm in the UK, Nvidia and Apple in the US) create chip designs
- Machine makers (ASML in the Netherlands) build the lithography tools
- Foundries (e.g. TSMC in Taiwan, Samsung in South Korea) manufacture chips
- Packaging and testing often happen in other countries again

KEY WORDS

SUPPLY CHAIN	All the steps, companies and countries involved in making and delivering a product.
CHIP SHORTAGE	A period (2020-2022) when demand for microchips was far greater than supply.
E-WASTE	Unwanted electrical and electronic devices.

CHIPS AND SECURITY

- Most of the world's most advanced chips are made by one company, TSMC, in Taiwan
- A disaster or conflict there could disrupt the whole world's technology
- The US, EU and others have spent tens of billions of pounds building new fabs
- The UK's strategy focuses on chip design, research and compound semiconductors

THE ENVIRONMENTAL COST

- Fabs use huge amounts of electricity and ultra-pure water
- Chemicals and gases must be handled carefully
- Short phone lifespans create e-waste
- Using devices longer and recycling them properly helps

ACTIVITY · 30 MINUTES · GROUPS OF 4

CHIP CRISIS SUMMIT

1. Your group represents a country at a "chip crisis summit": UK, USA, Taiwan or the Netherlands (use the fact cards).

2. Prepare: what is your country good at in the chip supply chain, and what do you depend on others for?

3. At the summit, each group has 1 minute to propose one action to make chip supplies safer and greener.

4. Vote on the best proposal and write a summary.

In your workbook, answer:

? What is your country's strength, and what does it depend on others for?

? Write the proposal your class voted for and explain why it would help.

Your workbook: Lesson 6 · tick each step off as you go

QUICK CHECK

1. What is a supply chain?

A A list of shops
B A chain for a bike
C A type of chip
D All the steps and companies involved in making a product

2. Which company is the only maker of EUV lithography machines?

A TSMC
B ASML
C Apple
D Arm

3. Which British company designs processor architectures used in most phones?

A Arm
B Rolls-Royce
C BT
D Tesco

4. Why did car factories stop production in 2021?

A A shortage of tyres
B A shortage of microchips
C A ban on cars
D Too many cars

QUICK CHECK: ANSWERS

1. **D: All the steps and companies involved in making a product**
   Chips have very complex supply chains.

2. **B: ASML**
   ASML is in the Netherlands.

3. **A: Arm**
   Arm is based in Cambridge.

4. **B: A shortage of microchips**
   Cars contain hundreds of chips.

2 MARKS · IN YOUR WORKBOOK

EXIT TICKET

Explain why governments see semiconductors as a national security issue.

Look back at today's success criteria: can you tick them all?

- ✓ I can describe the global semiconductor supply chain
- ✓ I can explain why governments see chips as a security issue
- ✓ I can describe environmental impacts of chip-making

END OF TOPIC

MICROCHIPS AND SEMICONDUCTORS

Revise each lesson in your workbook: the notes, key words and quick checks.
Then take the topic test when Mr Butler sets it.